WORKED SOLUTION

STEP I 1987, Question 9

Pure • Barycentric Coordinates

> **STEP I 1987, Question 9**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/87-s1-q9

The key idea

Moving from A towards B is a matter of adding a fraction of the vector $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$. Doing this once gives points on a segment. Doing it twice, first along AB and then from that point towards C, sweeps out the whole triangle. For the sketches, the trick is to rewrite each case so it looks like something we already understand: either a flip of the triangle case, or a point A plus multiples of two edge vectors.

Solution

Points on AB

A point P on the segment AB is reached from A by moving a fraction t of the way along $\overrightarrow{AB}$, with $0 \le t \le 1$ ($t = 0$ gives A and $t = 1$ gives B). So

$$\mathbf{p} = \mathbf{a} + t(\mathbf{b} - \mathbf{a}) = (1 - t)\mathbf{a} + t\,\mathbf{b}.$$

Put $\rho = 1 - t$ and $\sigma = t$. Then $\rho \ge 0$ and $\sigma \ge 0$ because $0 \le t \le 1$, and $\rho + \sigma = 1$. This is the required form. Conversely, every such pair ρ, σ gives a point of the segment, by taking $t = \sigma$.

Points on PC, and points inside the triangle

In exactly the same way, a point Q on the segment PC has position vector

$$\mathbf{q} = (1 - s)\mathbf{p} + s\,\mathbf{c}, \qquad 0 \le s \le 1.$$

Substituting $\mathbf{p} = \rho\mathbf{a} + \sigma\mathbf{b}$,

$$\mathbf{q} = (1 - s)\rho\,\mathbf{a} + (1 - s)\sigma\,\mathbf{b} + s\,\mathbf{c}.$$

Let $\lambda = (1 - s)\rho$, $\mu = (1 - s)\sigma$ and $\nu = s$. Each is a product of non-negative numbers, so $\lambda, \mu, \nu \ge 0$. Also

$$\lambda + \mu + \nu = (1 - s)(\rho + \sigma) + s = (1 - s) + s = 1.$$

Ask yourself Why does every point inside the triangle lie on some segment PC?

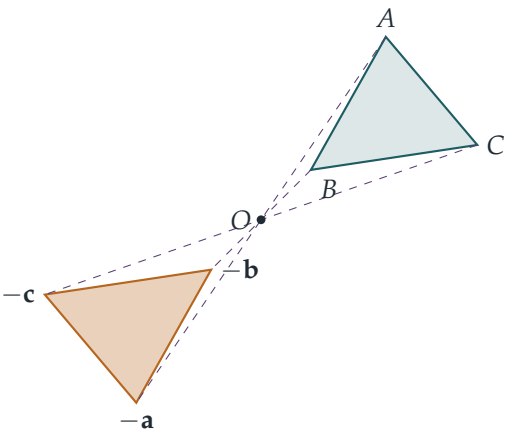
Take a point X inside the triangle (or on its boundary). If $X = C$ there is nothing to prove: take $s = 1$. Otherwise draw the line from C through X and carry it on beyond X. Because X lies inside the angle ACB, this line stays between CA and CB, so it must cross the opposite side AB, at a point P say. Since X is in the triangle, it lies between C and P, so X is on the segment PC. By the working above, $\mathbf{x} = \lambda\mathbf{a} + \mu\mathbf{b} + \nu\mathbf{c}$ with $\lambda, \mu, \nu \geq 0$ and $\lambda + \mu + \nu = 1$, as required.

Part (i)

Write $\lambda = -\lambda'$, $\mu = -\mu'$ and $\nu = -\nu'$. The conditions become $\lambda', \mu', \nu' \geq 0$ and $\lambda' + \mu' + \nu' = 1$, and the point is

$$\lambda\mathbf{a} + \mu\mathbf{b} + \nu\mathbf{c} = -\left(\lambda'\mathbf{a} + \mu'\mathbf{b} + \nu'\mathbf{c}\right).$$

The bracket runs over every point of the triangle ABC, by what we have just shown (with the converse from the first part, every such combination is in the triangle). So the region is the set of points $-\mathbf{x}$ with X in triangle ABC. In other words, it is the triangle with vertices $-\mathbf{a}$, $-\mathbf{b}$, $-\mathbf{c}$, together with its interior: the image of triangle ABC under a half-turn about the origin O.



The shaded amber triangle is the answer to (i). Each vertex is the reflection of a vertex of ABC through O, shown by the dashed lines.

Part (ii)

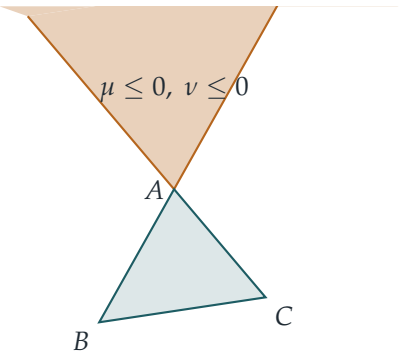
Now $\lambda = 1 - \mu - \nu$, so

$$\lambda\mathbf{a} + \mu\mathbf{b} + \nu\mathbf{c} = \mathbf{a} + \mu(\mathbf{b} - \mathbf{a}) + \nu(\mathbf{c} - \mathbf{a}).$$

We start at A and add multiples of $\overrightarrow{AB}$ and $\overrightarrow{AC}$. Writing $\mu = -m$ and $\nu = -n$ with $m, n \geq 0$, the point is

$$\mathbf{a} + m(\mathbf{a} - \mathbf{b}) + n(\mathbf{a} - \mathbf{c}), \qquad m \geq 0,\ n \geq 0.$$

The vector $\mathbf{a} - \mathbf{b}$ points along BA continued beyond A, and $\mathbf{a} - \mathbf{c}$ points along CA continued beyond A. Non-negative combinations of two non-parallel vectors fill exactly the angle between them (since $\mathbf{b} - \mathbf{a}$ and $\mathbf{c} - \mathbf{a}$ are not parallel, every point of the plane is $\mathbf{a} + m(\mathbf{a} - \mathbf{b}) + n(\mathbf{a} - \mathbf{c})$ for exactly one pair m, n). So the region is the infinite wedge with vertex A bounded by the extensions of BA and CA beyond A, boundary rays included. It is the angle vertically opposite to angle BAC. Note that λ isn't restricted in the question, but here it is automatically at least 1.



The amber wedge (continuing without end) is the answer to (ii). Its edges are the sides BA and CA extended through A. The teal triangle ABC is shown for reference.

Checks

- Vertices: in (ii), $(\lambda, \mu, \nu) = (1, 0, 0)$ gives A itself, the tip of the wedge; $(2, -1, 0)$ gives $2\mathbf{a} - \mathbf{b}$, the reflection of B in A, which lies on one edge. In (i), $(-1, 0, 0)$ gives $-\mathbf{a}$, a vertex of the reflected triangle.
- Centroids: in (i), $\lambda = \mu = \nu = -\frac{1}{3}$ gives minus the centroid of ABC, which is the centroid of the reflected triangle, as it should be.
- Numerically, a computer took 20,000 random triangles and random coefficients satisfying each set of conditions. It confirmed that every point from (i) lies in the triangle $(-\mathbf{a}, -\mathbf{b}, -\mathbf{c})$, that every point from (ii) has non-negative coordinates along $\mathbf{a} - \mathbf{b}$ and $\mathbf{a} - \mathbf{c}$ from A, and that every point of ABC splits as a point of some segment PC.

Common slips

- In (i), drawing a region near ABC. The coefficients sum to -1, not 1, so the region is the half-turn of ABC about the origin and depends on where O is.

- In (ii), drawing only a triangle. There is no lower bound on μ and ν, so the region is unbounded.
- Shading the angle next to BAC (for example between AB and CA extended). That would need one of μ, ν positive.

Beyond STEP

The numbers (λ, μ, ν) with $\lambda + \mu + \nu = 1$ are the *barycentric coordinates* of a point with respect to the triangle. When the sum is 1 the point doesn't depend on where the origin is: shifting every position vector by $\mathbf{d}$ shifts $\lambda\mathbf{a} + \mu\mathbf{b} + \nu\mathbf{c}$ by $(\lambda + \mu + \nu)\mathbf{d} = \mathbf{d}$ too. That is why the answer to (ii) is a genuine geometric region, while (i) depends on O. The triangle itself is the *convex hull* of A, B, C, an idea at the heart of linear programming and convex optimisation. The signs of the barycentric coordinates split the plane into seven regions, of which (ii) is one. For a first-year view of these affine ideas, see Strang, *Introduction to Linear Algebra*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.